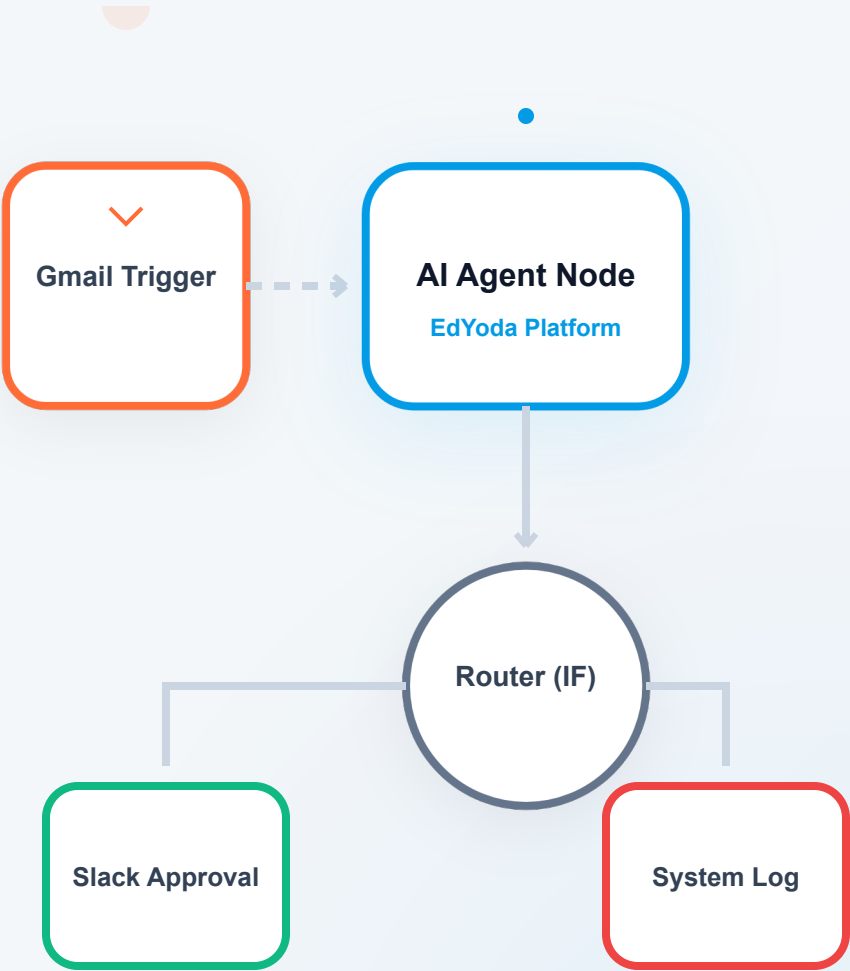


Workflow Automation with n8n

Wiring Agents into Business Workflows

OUTCOME-BASED LEARNING GOAL

Build a working n8n workflow that triggers an EdYoda agent, routes output through conditional logic, includes human approval, and monitors execution — without coding.



The Bottleneck: Siloed Systems & Manual Glue

Why manual AI orchestrations choke scaling enterprise operations.

THE BOTTLENECK

AI agents can reason, but they cannot act alone. Human staff are acting as the "human glue" between tools.

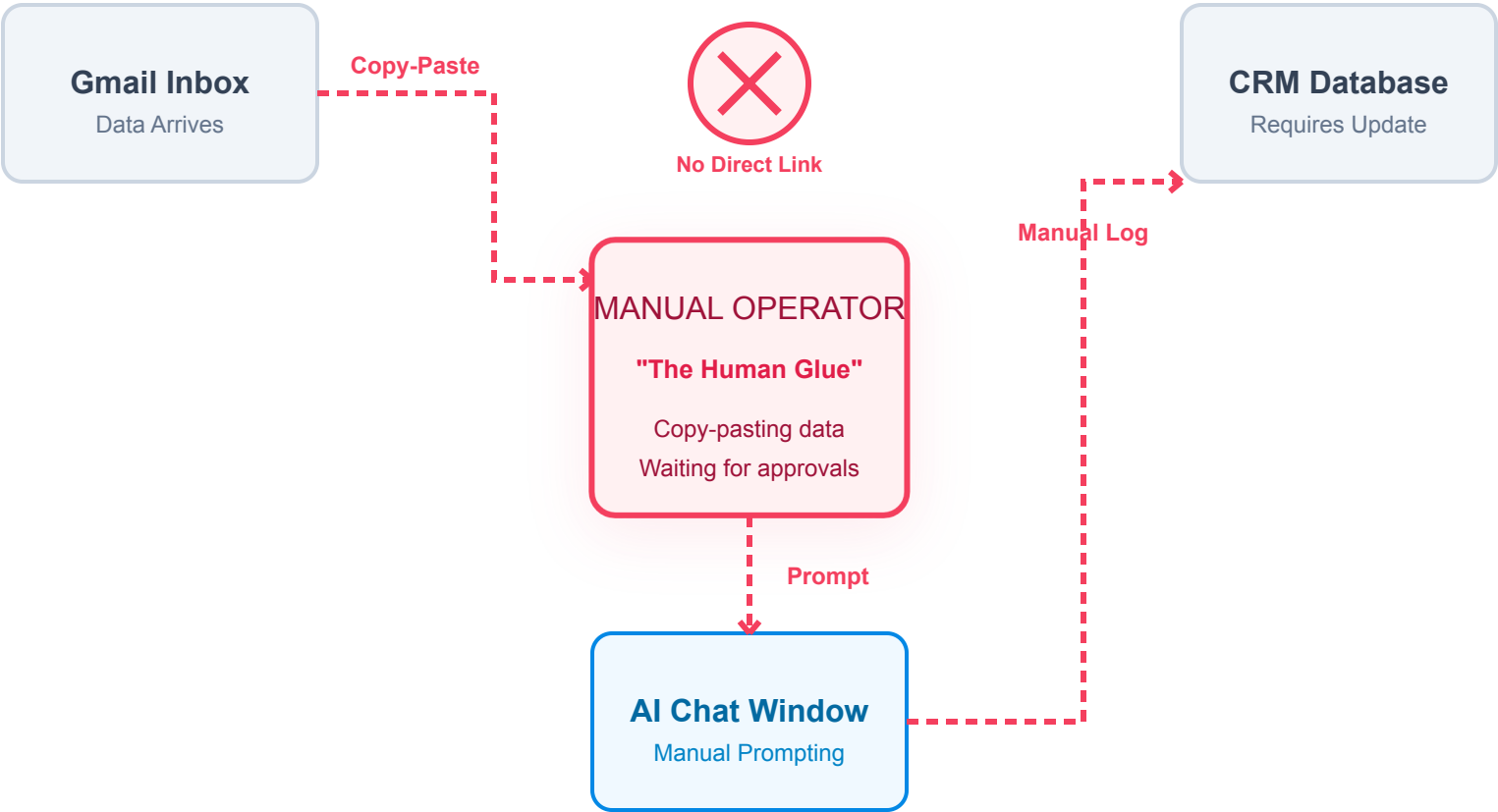
KEY OPERATIONAL PAIN POINTS

- Manual data copy-pasting between separate legacy platforms.
- Valuable AI intelligence trapped inside disconnected browser tabs.
- Unstructured handoffs causing massive delays in critical approvals.
- No clear execution tracking or centralized audit logs.

BUSINESS Metaphor

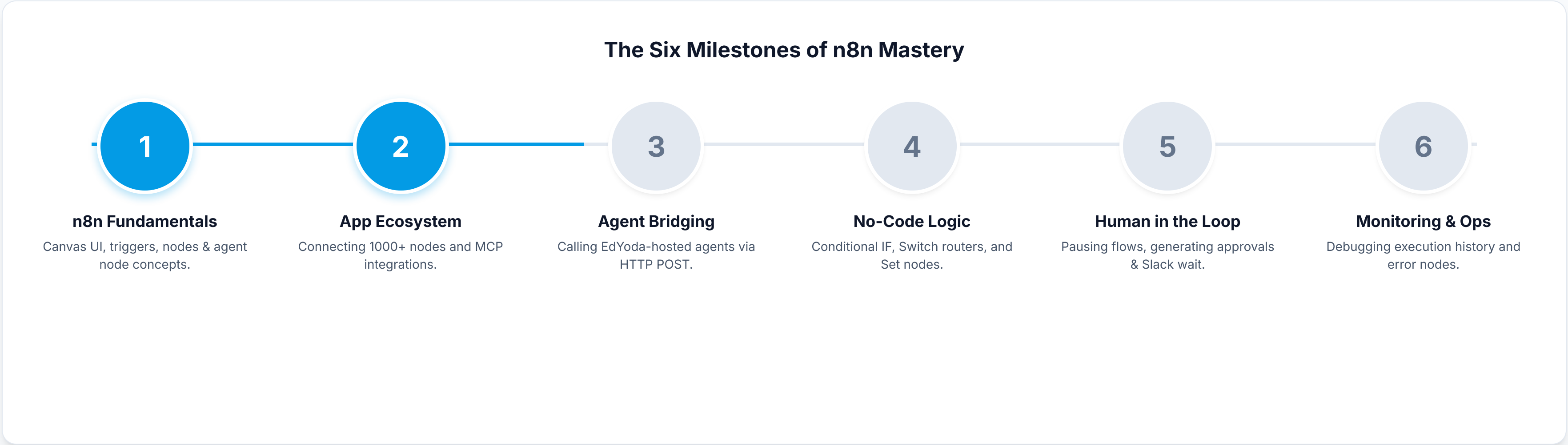
"Buying a hyper-car (LLM Brain) but having to push it manually (data entry) down the highway."

Operational Chokepoint: The "Human Glue"



Course Roadmap: Wiring Agents into Workflows

A step-by-step progression from core builder fundamentals to enterprise deployment.



ACTIVE DISCUSSION: MAPPING EVERYDAY WORKFLOWS

"Think of one repetitive process in your role that involves more than 2 tools or apps."

How many manual steps does it take today? How many decisions do you make? We will discover how many nodes would it take to automate this entire process in n8n.

Visual Workflow Builder & AI Agent Node

Understanding the central nervous system of enterprise automation.

THE CENTRAL NERVOUS SYSTEM ANALOGY

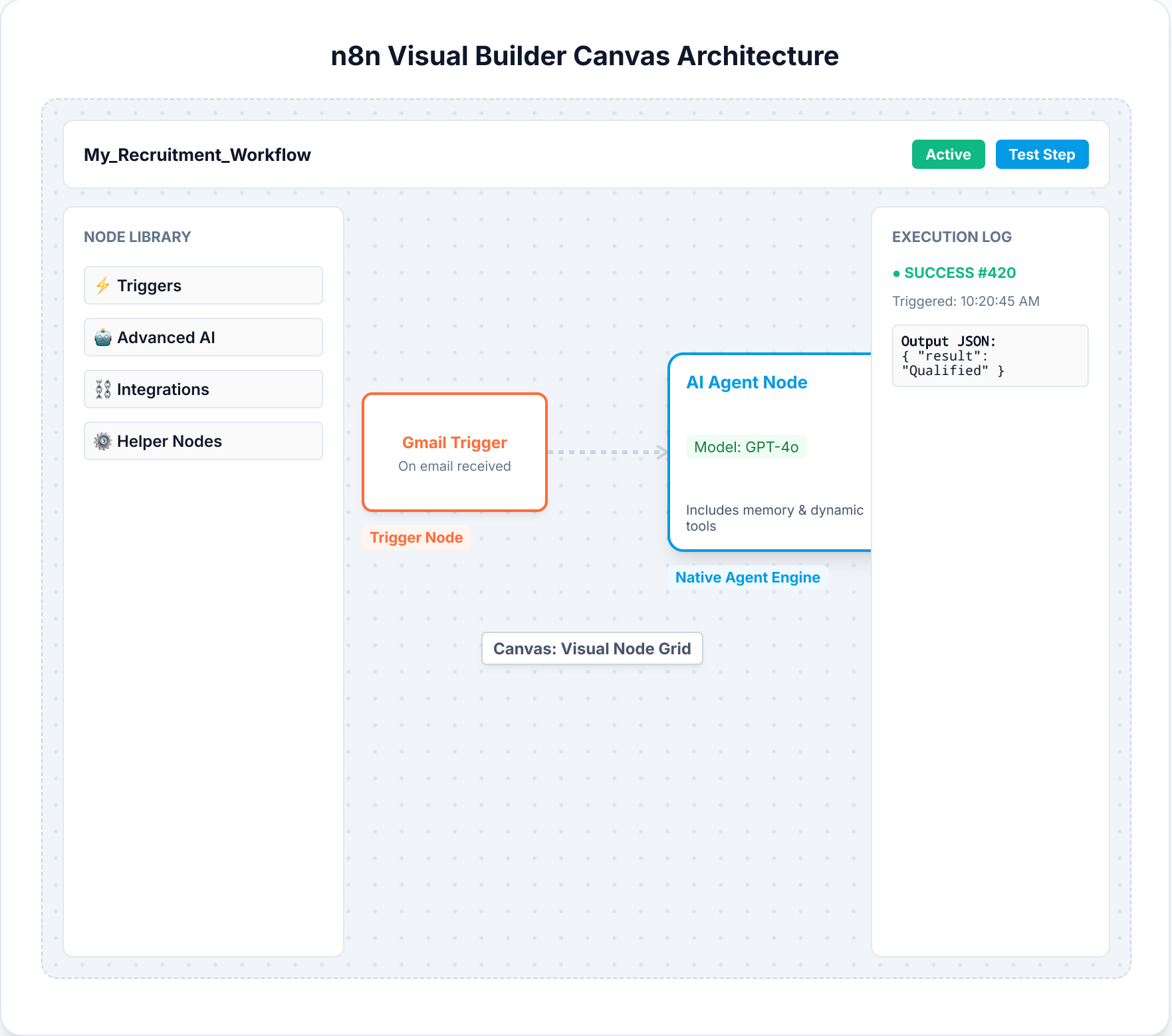
n8n is the nervous system. Your AI agents are the brains. n8n connects the brain to every application, tool, and database.

CORE N8N TERMINOLOGY

- **Workflow:** Connected sequence of nodes executing automatically. (Like flowcharts)
- **Node:** Single unit of specialized work or tool connection.
- **Trigger:** Event initiating workflow run. (E.g. Webhook, Schedule)
- **Execution:** Single workflow run logging input, output, and status.
- **Credentials:** Secure, reusable API keys saved once across canvas.

AI AGENT NODE INTEGRATION

n8n's native Agent node lets you load LLMs, configure system prompts, plug in tools, and establish persistent session memory directly inside the visual canvas interface.



Step-by-Step: Assembling a Workflow

A live breakdown of a 4-node production automation sequence.

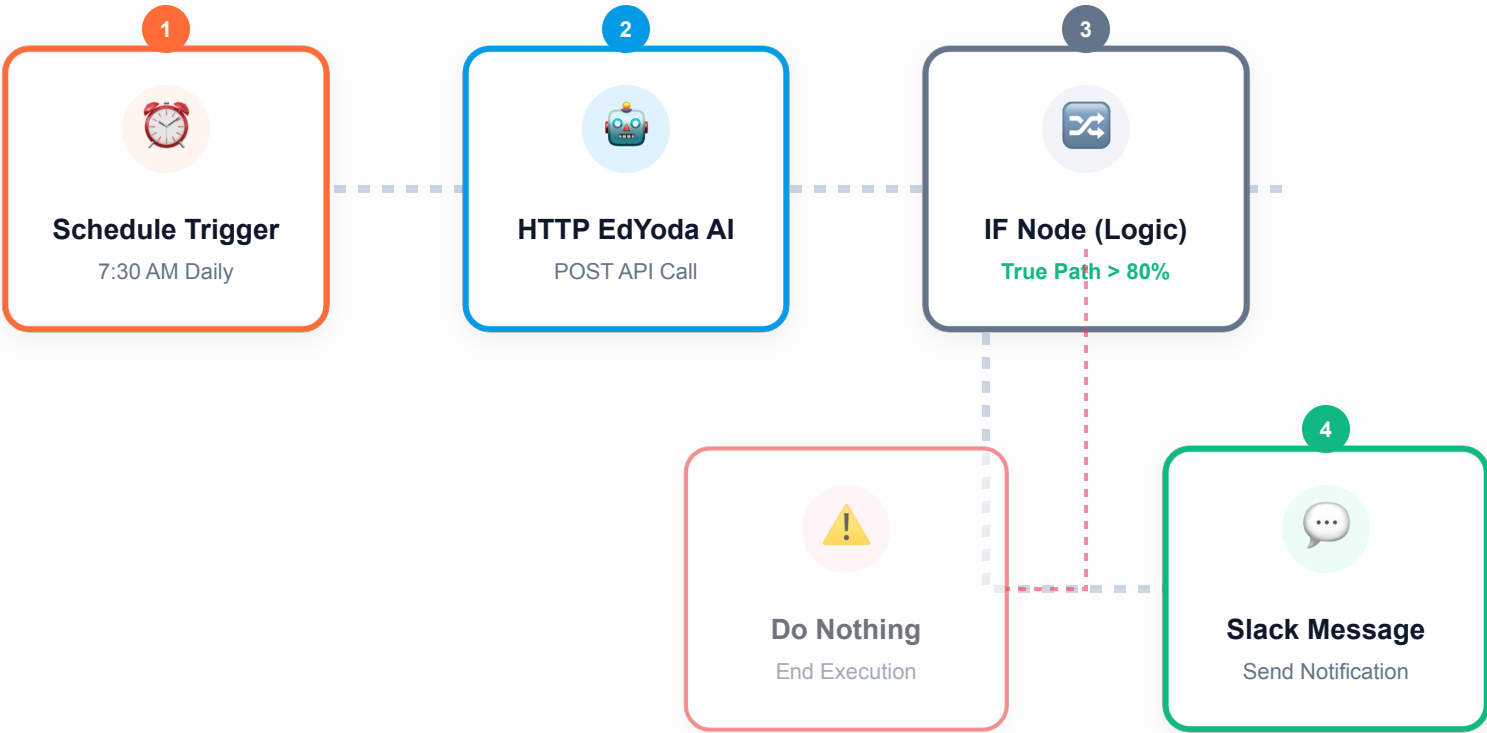
Workflow Assembly Sequence

- 1 Schedule Trigger Setting**
Triggers automatically at 7:30 AM every business day.
- 2 Call EdYoda AI Agent**
Calls external agent endpoint passing dynamic contextual data.
- 3 Route with IF Condition**
Inspects AI output JSON; checks if confidence score exceeds 80%.
- 4 Send Instant Slack Notification**
Pushes structured briefing directly into targeted HR Slack channels.

INSTRUCTIONAL TAKEAWAYS

- Drag and connect nodes sequentially to assemble automation workflows.
- Configure specific credentials in credentials tab for each app.
- Map data outputs visually without writing JavaScript transformation code.

Step-by-Step Execution Path



1000+ App Connections & MCP Integration

Unifying disparate databases, CRMs, chat engines, and external AI protocols.

UNIVERSAL POWER ADAPTERS

Pre-built nodes are like universal power adapters. Plug in, authenticate once, and immediately start exchanging formatted payload data safely.

KEY APP INTEGRATION STRENGTHS

- **Native MCP Support:** One protocol connects agent to multiple external resources.
- **Zero-Code Auth:** Standardized OAuth2 credentials managed securely on centralized canvas.
- **Automatic Schema Parsing:** Dynamic schema engine pulls field names from APIs automatically.
- **Pre-built Library:** 90% of enterprise tools covered natively out of box.

IN PRACTICE: RECRUITMENT PIPELINE MAP

Google Forms (Trigger) → Gmail (Read Resume) → EdYoda AI (Screen Candidate) → Google Sheets (Log) → Calendly API (Schedule) → Slack (Alert HR)

n8n Central Integration Hub Architecture



Anatomy of Live Node Configurations

Step-by-step panel walkthrough for Gmail, HTTP Request, and Google Sheets nodes.

Gmail Node Setup

Monitors incoming resumes or inquiries.

Resource

Message

Event

On Email Received

Filters

label: inbox, has:attachment

- Detects new emails with specified triggers instantly.
- Extracts PDF files from incoming messages.

HTTP Request Setup

Calls EdYoda-hosted reasoning agent.

Method

POST

URL

api.edyoda.com/v1/agents/run

Authentication

Header Auth (API Token)

- Injects files from the Gmail step dynamically.
- Receives candidate review summaries back immediately.

Google Sheets Setup

Logs applicant evaluation outcomes.

Resource

Spreadsheet Document

Operation

Append Row

Sheet File

HR_Candidate_Tracker

- Pushes formatted JSON outputs to tracker automatically.
- Locks records instantly to prevent data loss.

The HTTP Request Node Pattern

Bridging workflow automation triggers with advanced AI logical reasoning.

THE TELEPHONE ANALOGY

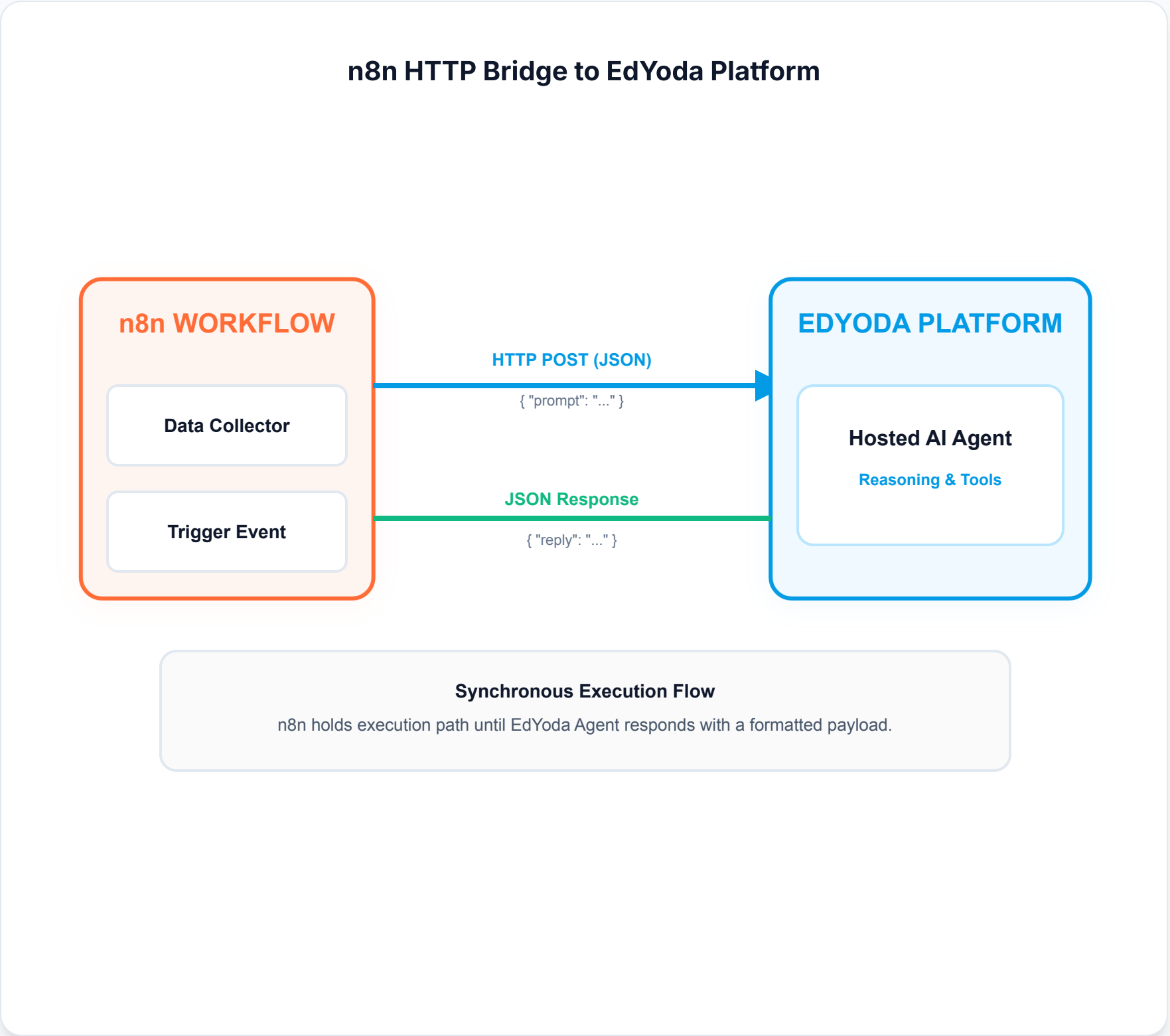
The HTTP node is a phone. Dial the agent (Endpoint), speak your message (Request Body), and listen to results.

STEP-BY-STEP API BRIDGING

- **Step 1:** Retrieve unique endpoint URL from EdYoda Deployment panel.
- **Step 2:** Set node method to POST and add Authorization header.
- **Step 3:** Build request body injecting data from preceding trigger nodes.
- **Step 4:** Parse returned JSON payload directly into downstream logic routers.

DYNAMIC CONTEXT INJECTION

Extract email bodies or database row entries on-the-fly and pass them as system variables inside the agent's reasoning prompt at execution time.



HTTP Request Node Walkthrough

Annotated API payload parameters and typical calling integration pattern schemas.

API Request Configuration Payload

```
// HTTP Request POST to EdYoda API
{
  "agentId": "recruitment_screener_v2",
  "variables": {
    "candidate_name": "{{ $json.email_sender }}",
    "resume_text": "{{ $json.parsed_pdf_text }}"
  },
  "parameters": {
    "temperature": 0.2,
    "confidence_floor": 0.8
  }
}
```

Three EdYoda Calling Patterns

- 1. Synchronous Call Pattern**
Wait for the agent's evaluation reply before continuing to subsequent steps.
- 2. Asynchronous Fire-and-Forget**
Trigger agent process and exit immediately. Retrieve results via subsequent execution hook.
- 3. Loop Batch Calling**
Loop list items (e.g. daily leads) calling EdYoda once for each item.

 COMMON CONFIGURATION TRAP

Missing **Content-Type: application/json** in HTTP headers will cause API endpoint validation errors. Always verify under "Headers" panel!

PARSING EXPRESSIONS

Extract model's text field using dynamic expressions:
`{{ $json.body.response.text }}`

Conditional Logic, Branching & Trees

Constructing smart routing frameworks based on dynamic runtime variables.

THE TRAFFIC LIGHT ANALOGY

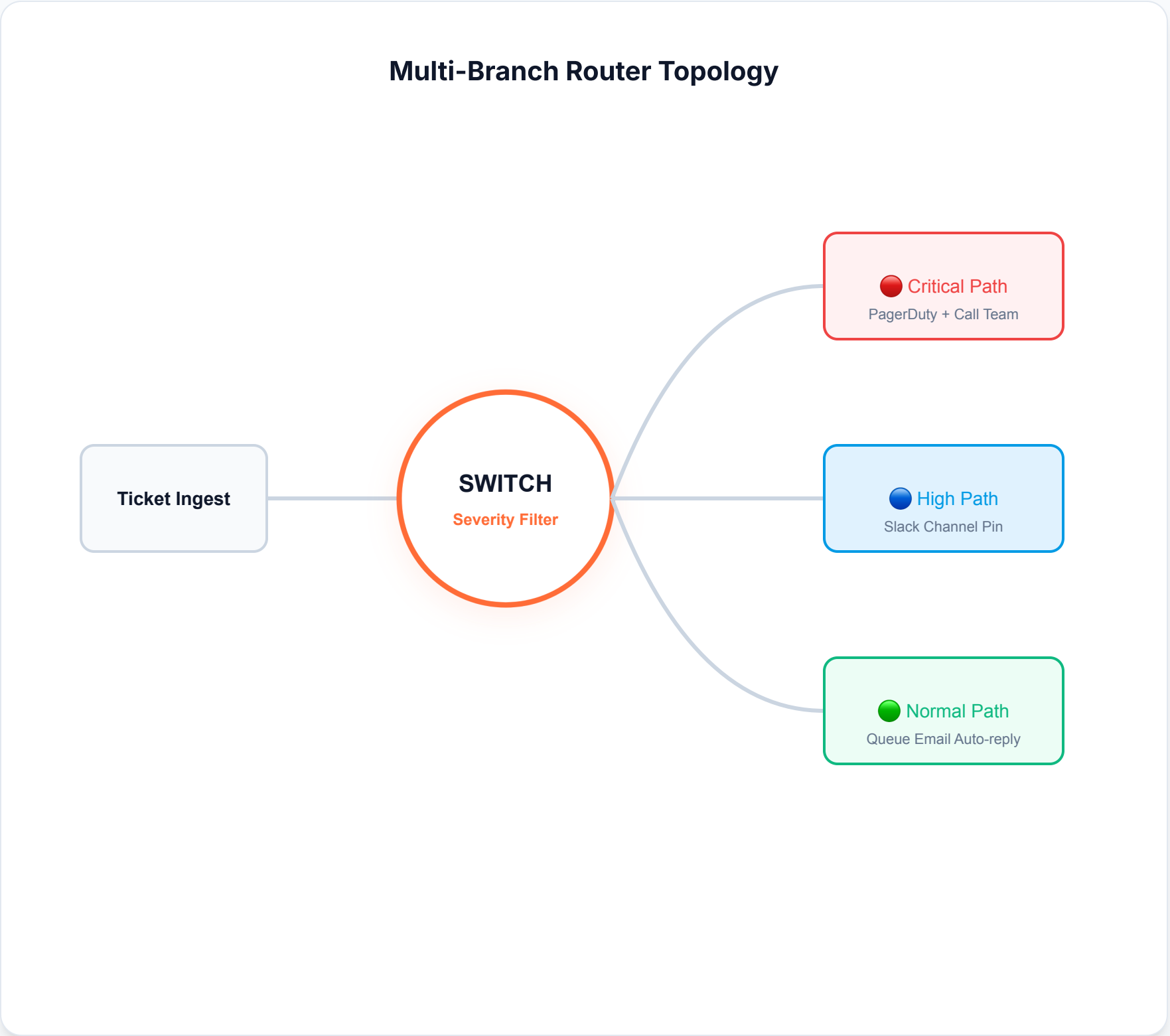
The IF node acts as a traffic light. Green allows true path execution; Red halts or diverts to alternate paths.

VISUAL LOGIC OPERATORS

- **IF Node:** Evaluates boolean statements to split execution into binary paths.
- **Switch Node:** Multi-way junction routing based on specific categories or scores.
- **Set Node:** Dynamically injects or restructures fields on active schemas.
- **Merge Node:** Recombines independent branching workflows back into singular streams.
- **Filter Node:** Blocks non-compliant payload items while letting high-score entries pass.

SWITCH NODE IN PRACTICE

Route inbound support tickets by severity: Critical triggers PagerDuty alerts, High sends Slack, and Low queues normal emails.



Case Study: Inbound Intent Email Router

A live production deployment diagram automatically sorting support requests by semantic intent.

Routing Pipeline Operations

Ingestion (Gmail Trigger)

Pulls raw email subject lines and body payloads sequentially.

Intent Classification (AI)

EdYoda classifier maps request semantic intent into high-level groups.

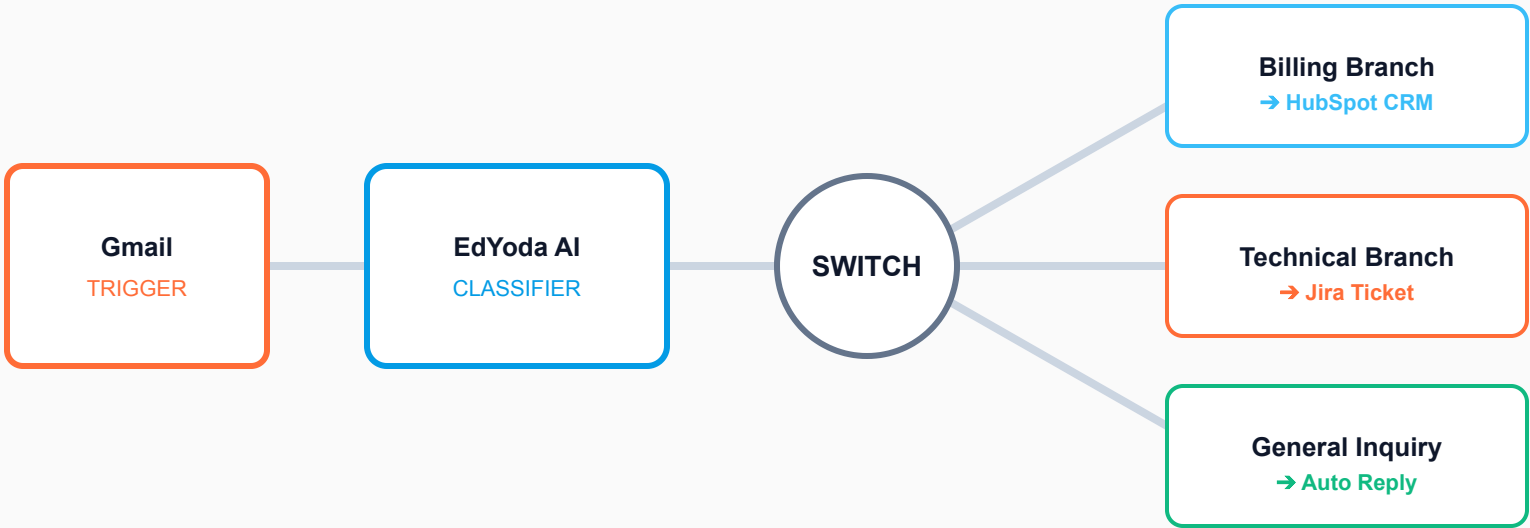
Route Verification (Switch)

Splits into Billing, Technical, or General operational queues.

THOUGHT CHALLENGE

"Take your chosen process. Draw the logical decision points. Where does the data path split? Map each split to an n8n IF or Switch node."

The Inbound Route Map



Human-in-the-Loop Nodes: Pausing Flows

Knowing when to inject critical human validation checkpoints into automated runs.

THE SAFETY VALVE ANALOGY

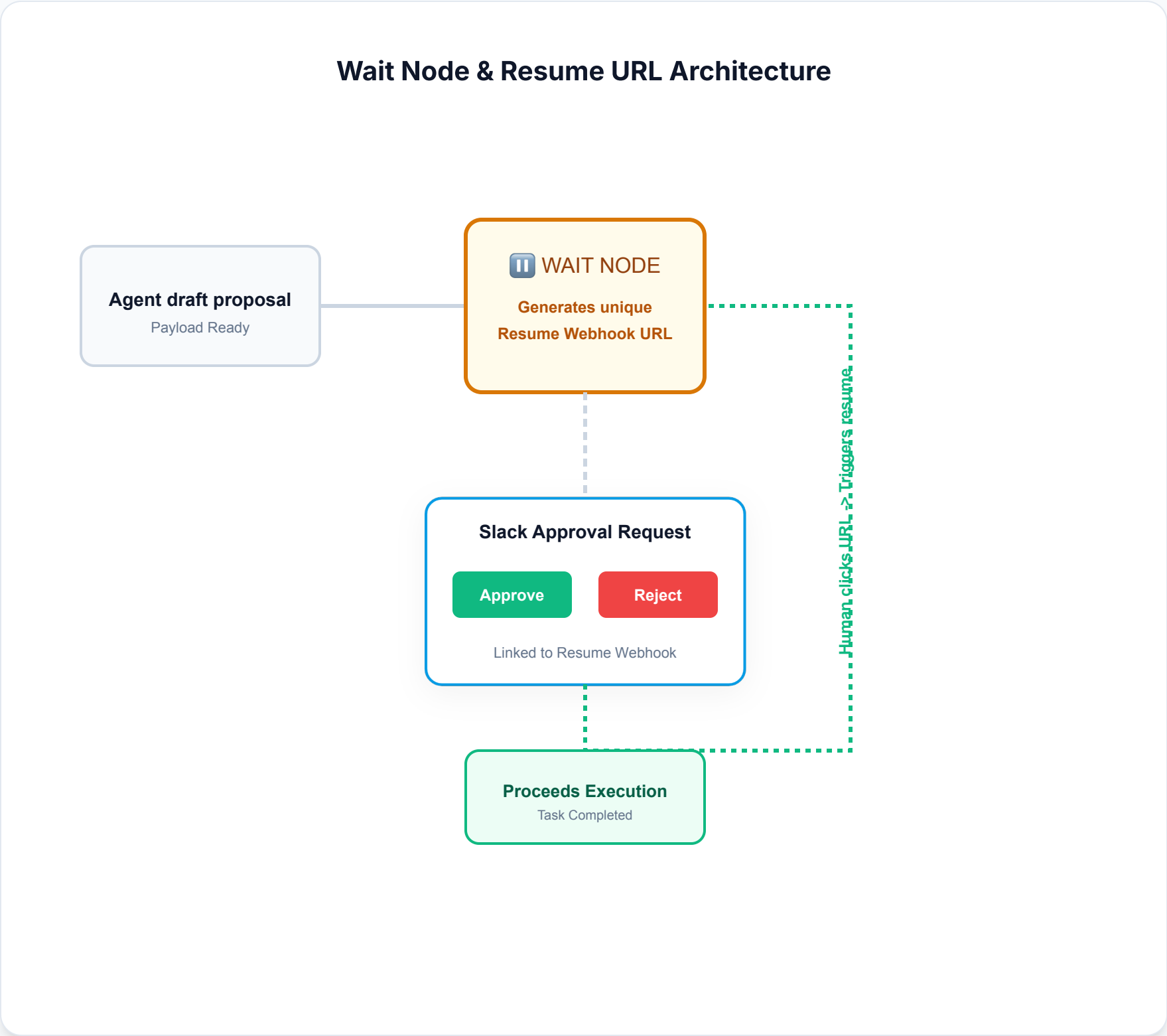
Human checkpoints are safety valves. Full automation is dangerous. Stop the flow to check risk parameters.

FIVE SITUATIONS REQUIRING HUMAN APPROVALS

- **Financial Transactions:** High value transfers exceeding defined expense threshold criteria.
- **External Comms:** Broadcast-level outgoing announcements generated automatically by LLM brains.
- **Hiring Decisions:** Final candidates selected, hired, or rejected by scoring nodes.
- **Legal Actions:** Approving dynamic SLA modifications before filing official repository commits.
- **Irreversible Steps:** Actions like database wiping which cannot be restored retroactively.

N8N WAIT NODE ARCHITECTURE

The **Wait Node** suspends active runtime execution, generating a distinct webhook resume URL. Execution continues only upon human interactive click.



Case Study: Recruitment Human Validation

A high-stakes pipeline protecting candidate experience through manual oversight checkpoints.

Workflow Architecture

1. AI Candidate Evaluation

EdYoda agent screens high-stakes resumes and drafts structured reviews.

2. Wait Hook Execution

Workflow halts and issues Slack button linked to dynamic callback endpoints.

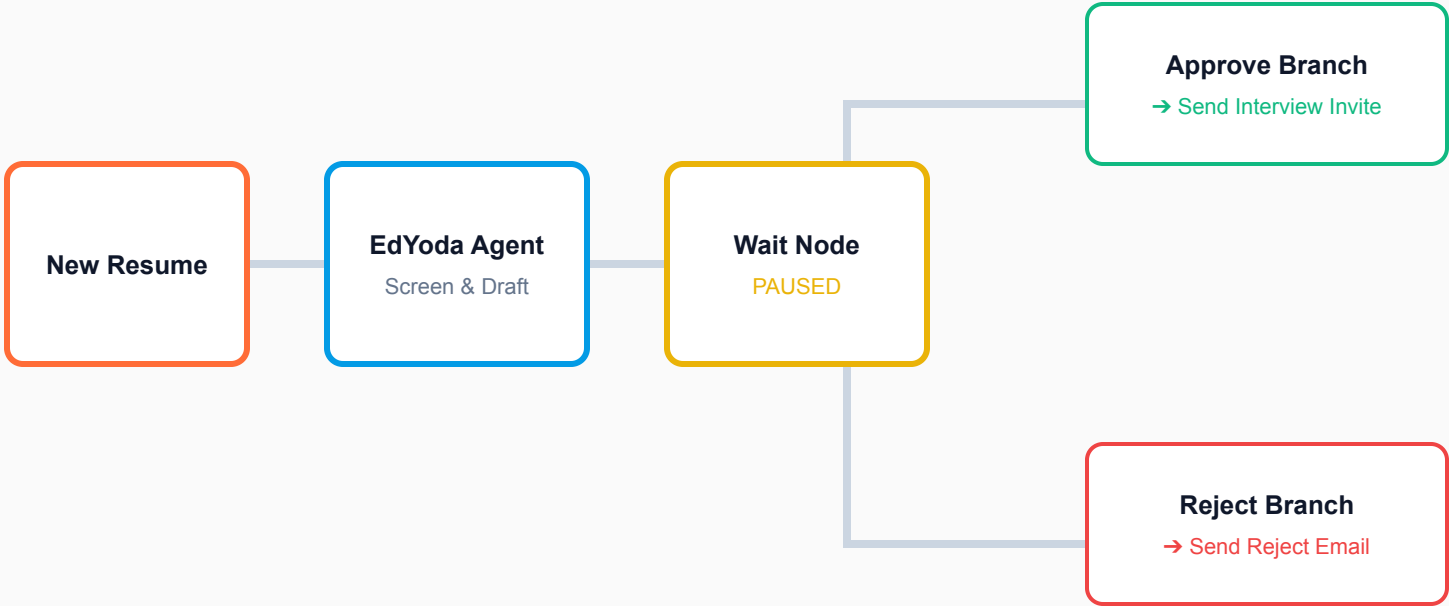
3. Logging & Validation Audit

Log reviewer username and specific approval decision into sheet archives.

CONTRACT REVIEW PATTERN

Agent drafts contract amendment → Pauses → Sends Slack to legal with Approve / Request Changes / Reject buttons → On Approve, workflow sends contract to client automatically.

Production Recruiter Layout



Visual Execution Logs & Alert Systems

Analyzing runtime trends, tracing error nodes, and optimizing latency.

THE AUDIT LOG SUPERPOWER

Every workflow run logs precise inputs and outputs per node. Tracing errors doesn't require guessing; the variables speak for themselves.

4-STEP DEBUGGING WORKFLOW

- **Step 1: Identify:** Locate failed red execution state inside history sidebar.
- **Step 2: Isolate:** Click failed execution to reveal erring node immediately.
- **Step 3: Diagnose:** Verify input parameters; search for malformed JSON schemas.
- **Step 4: Fix:** Correct setup, and trigger "Execute from node" rerun.

IN PRACTICE: METRICS TO TRACK

Daily executions count • Average run latency • Node failure rate • AI output accuracy score • Human approval response time.

n8n Visual Execution Tracing Window

EXECUTION HISTORY

Workflow ID: active_recruiter_01

SUCCESS10:20 AMID: #10294 • Webhook

FAILED10:15 AMID: #10293 • Gmail

SUCCESS09:40 AMID: #10292 • Webhook

ERRORNode: "Call EdYoda Agent (HTTP)"

ERROR: Bad Request (Status 400)

Input received:

{ "candidate": null, "text": "resume..." }

Reason:

"candidate_name" variable resolved to empty string (null).

SOLUTION IN n8n CANVAS:

Change expression in previous step: replace \$json.sender with fallback string \$json.sender_name || 'Anonymous'. Rerun execution!

Daily Briefing Agent Architecture

A fully automated daily intelligence delivery pipeline powered by n8n.

Daily Briefing Agent

Runs every morning at 7:30 AM to summarize and prioritize your day.

Human-in-the-Loop Pattern: Optional checkpoint. If an email is flagged as urgent requiring an external reply, the workflow pauses and requests confirmation on draft replies.

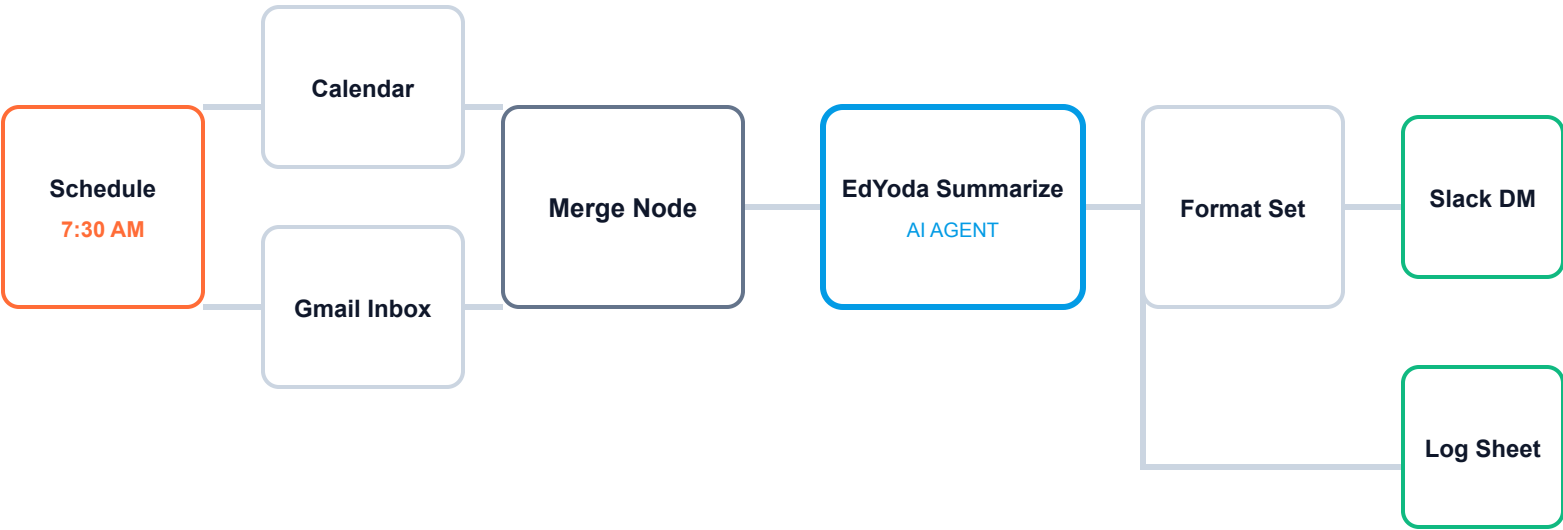
INPUTS (AUTO-INGESTED)

-  **Google Calendar:** Pulls scheduled events and meetings for today.
-  **Gmail Node:** Scans top 10 unread inbox emails (last 12 hours).
-  **Google Sheets:** Loads dynamic high-priority project focus codes.

STRUCTURED SLACK OUTPUTS

-  **Today's Meetings:** Times, attendees, and key agenda preparation notes.
-  **Email Triage:** Top 3 emails requiring actions with draft responses.
-  **Priority Alert:** Warns when any contact overlaps with flagged codes.

n8n Pipeline Architecture



Mastery Recap & The Road Ahead

Synthesizing workflow automation strategies and preparing for end-to-end sandbox deployment cases.

Strategic Synthesis



n8n Fundamentals

Visual workflow builder canvas design.



1000+ Apps Integration

Pre-built universal integration app layer.



HTTP Agent Bridge

Call any EdYoda hosted agent.



Conditional Logic Routing

Branch and filter without code.



Human-in-the-Loop

Pause flows for validation triggers.



Trace History

Debug and optimize performance visually.

NEXT TOPIC BRIDGE

Apply the full stack to two real business problems: Recruitment and Customer Support. That is where everything comes together.

INTERACTIVE REFLECTION PROMPT

"Map one process in your organization to an n8n workflow. What is the trigger? Which apps connect? Where does an EdYoda agent get called? Where is human oversight needed? Draw your map."